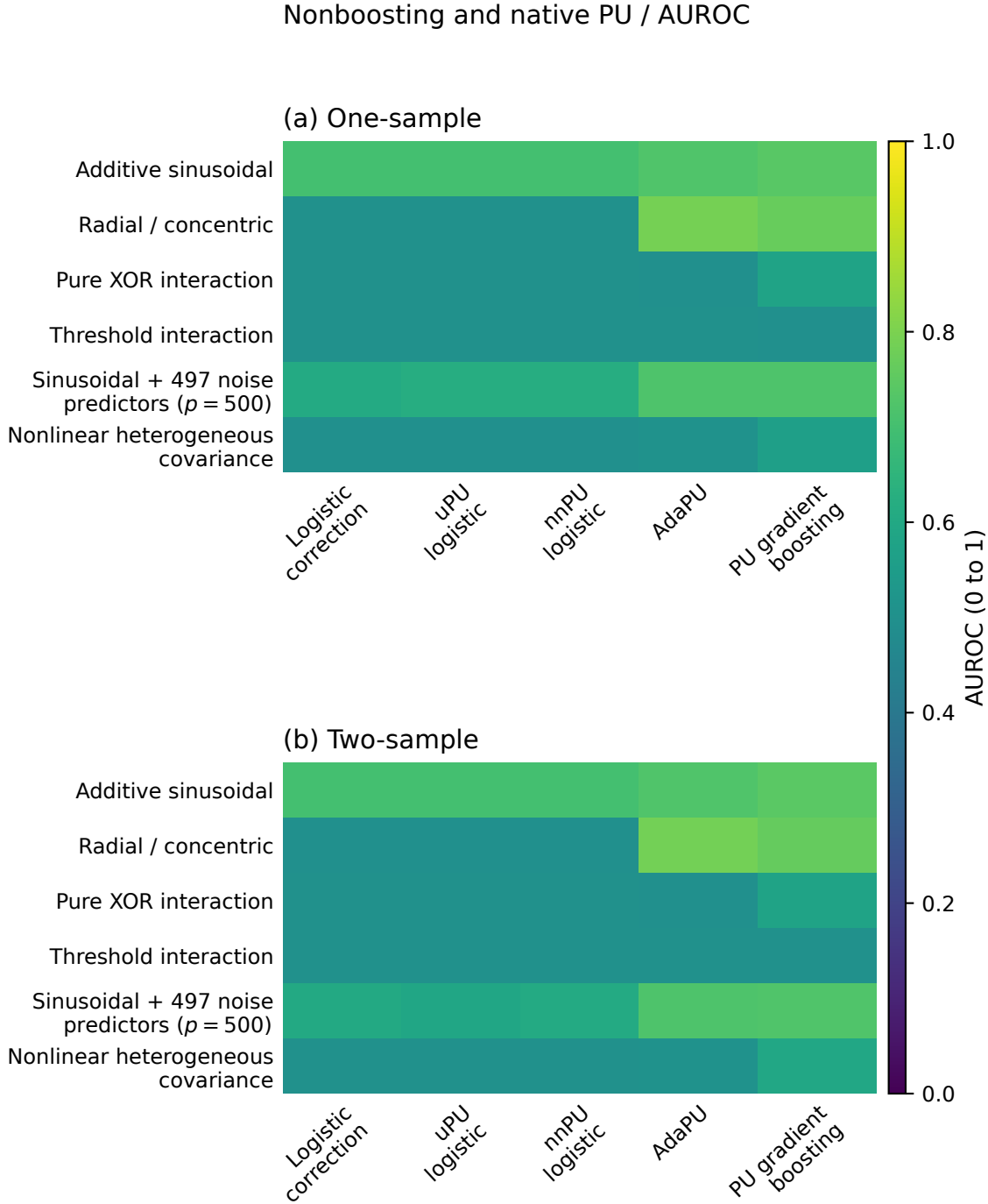
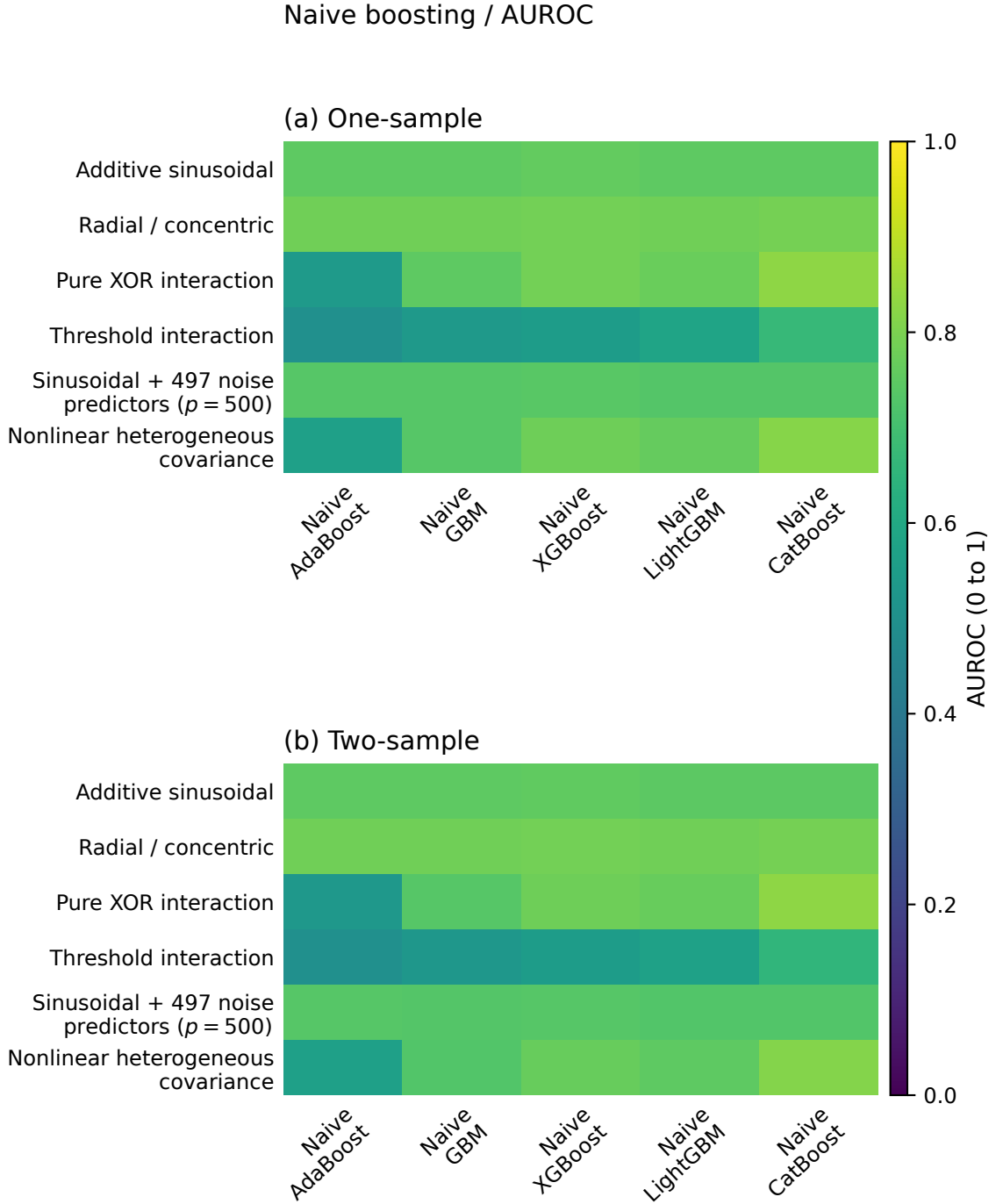
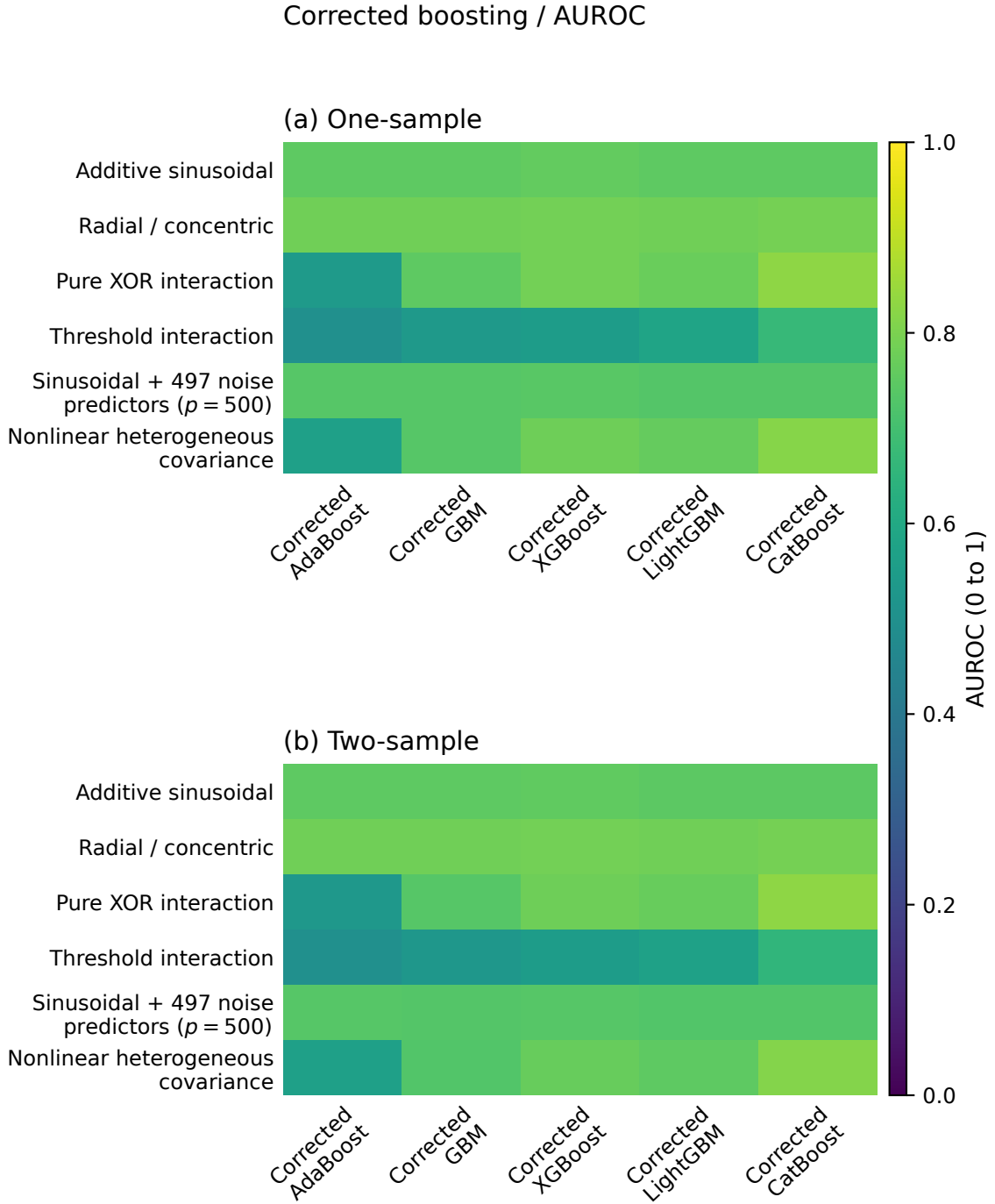
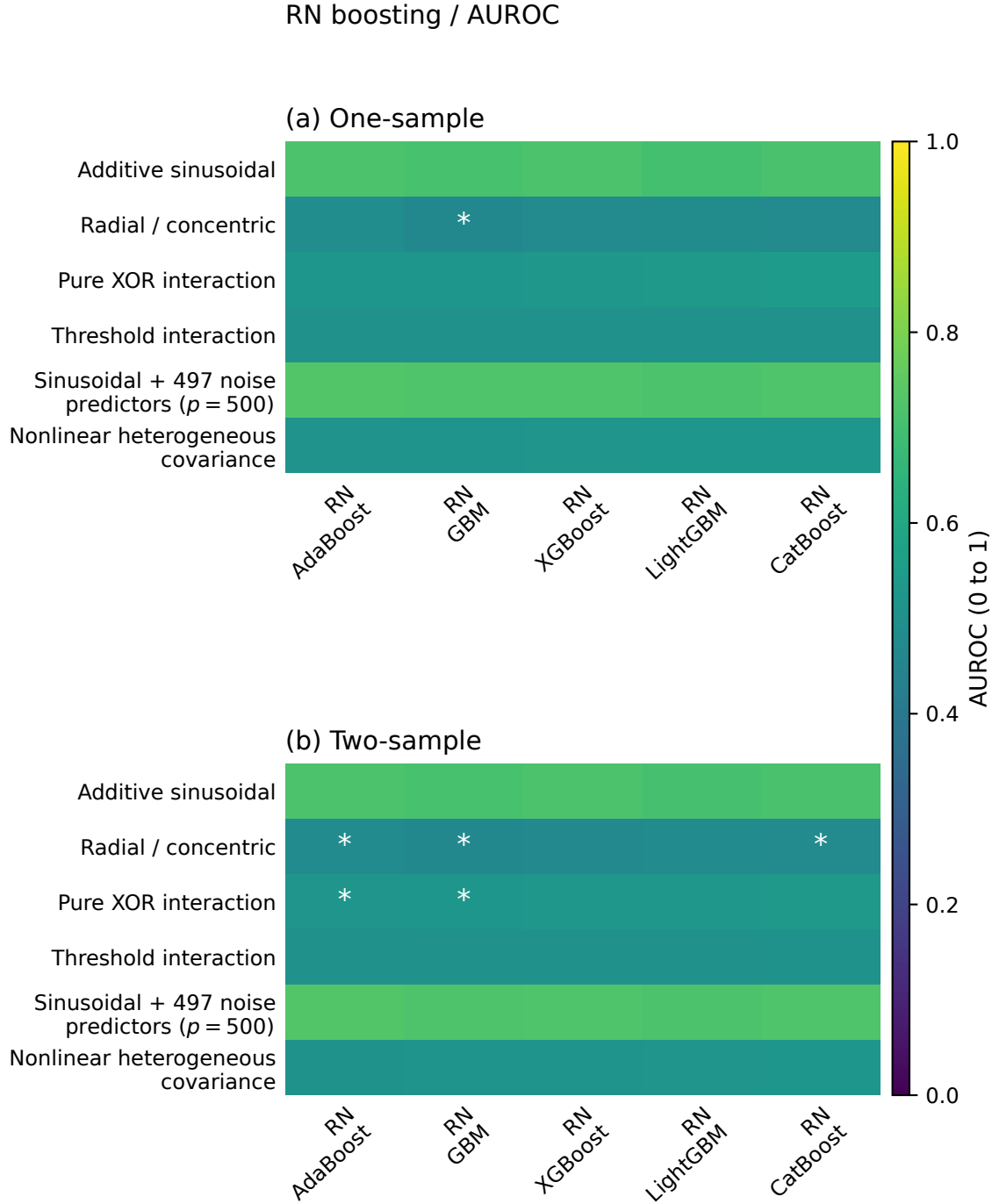
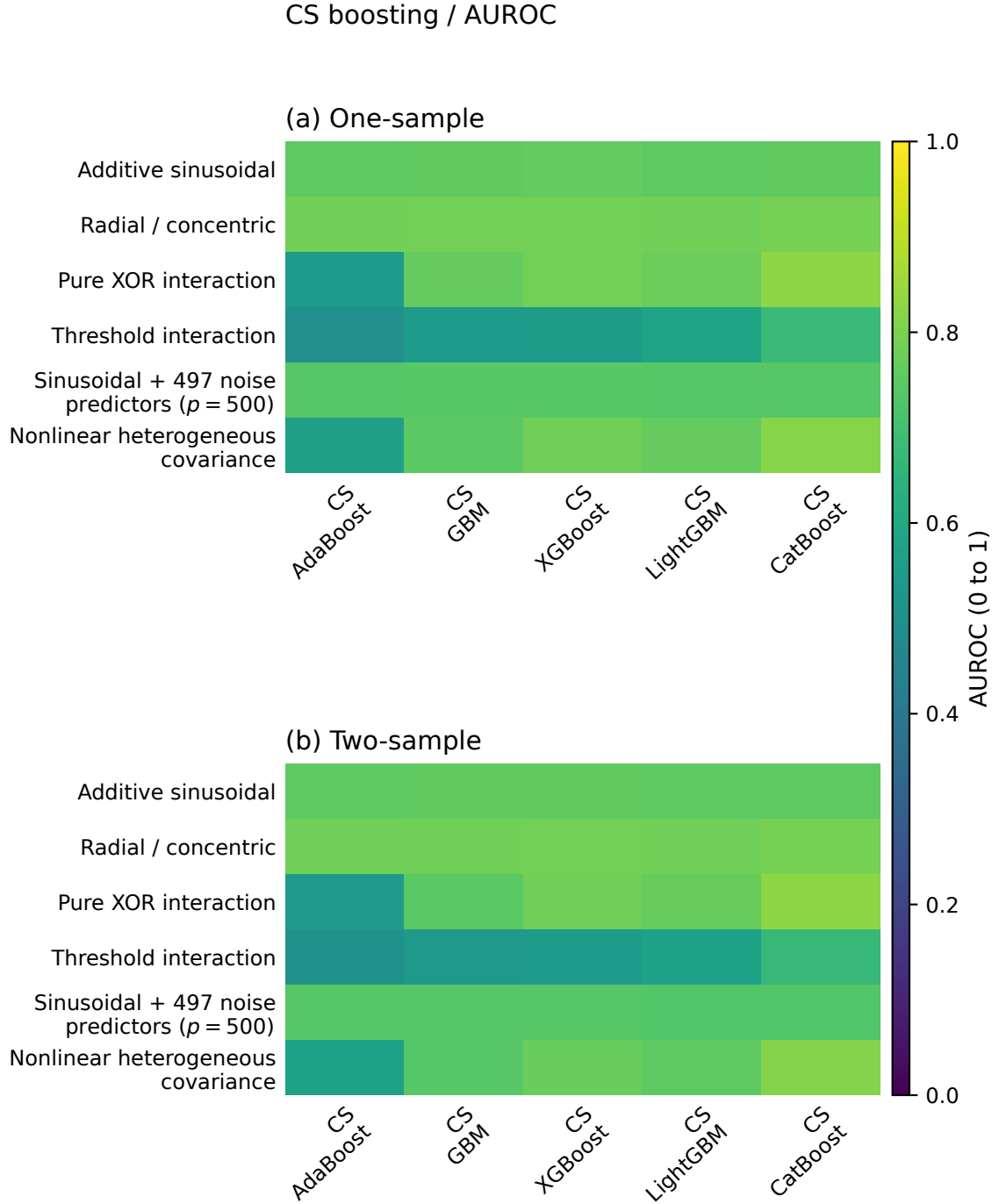
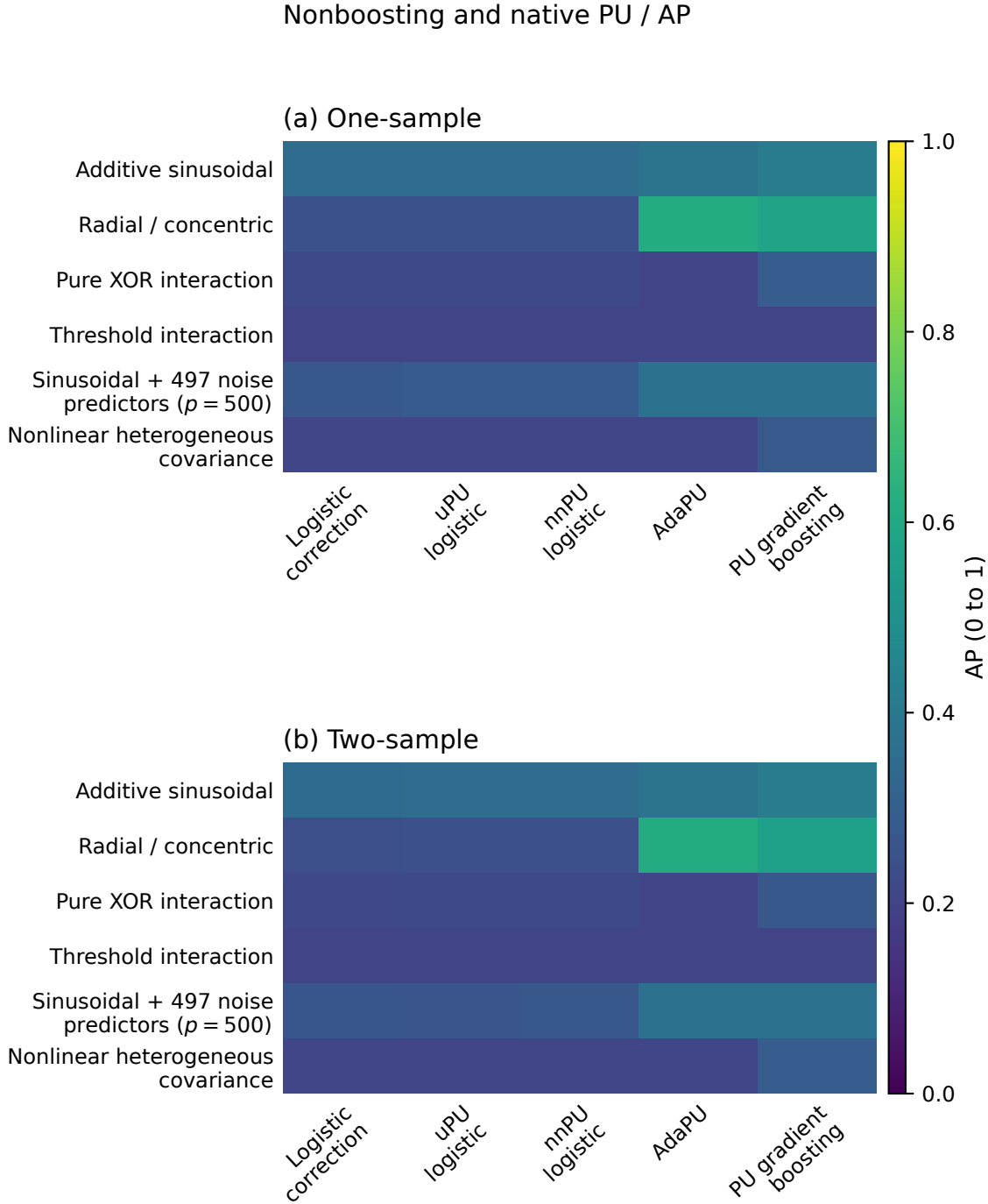

Nonboosting and native PU / AUROC
(a) One-sample
Additive sinusoidal
Radial / concentric
Pure XOR interaction
Threshold interaction
Sinusoidal + 497 noise predictors ($p = 500$)
Nonlinear heterogeneous covariance
Logistic correction
uPU logistic
nnPU logistic
AdaPU
PU gradient boosting
(b) Two-sample
Additive sinusoidal
Radial / concentric
Pure XOR interaction
Threshold interaction
Sinusoidal + 497 noise predictors ($p = 500$)
Nonlinear heterogeneous covariance
Logistic correction
uPU logistic
nnPU logistic
AdaPU
PU gradient boosting
AUROC (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0

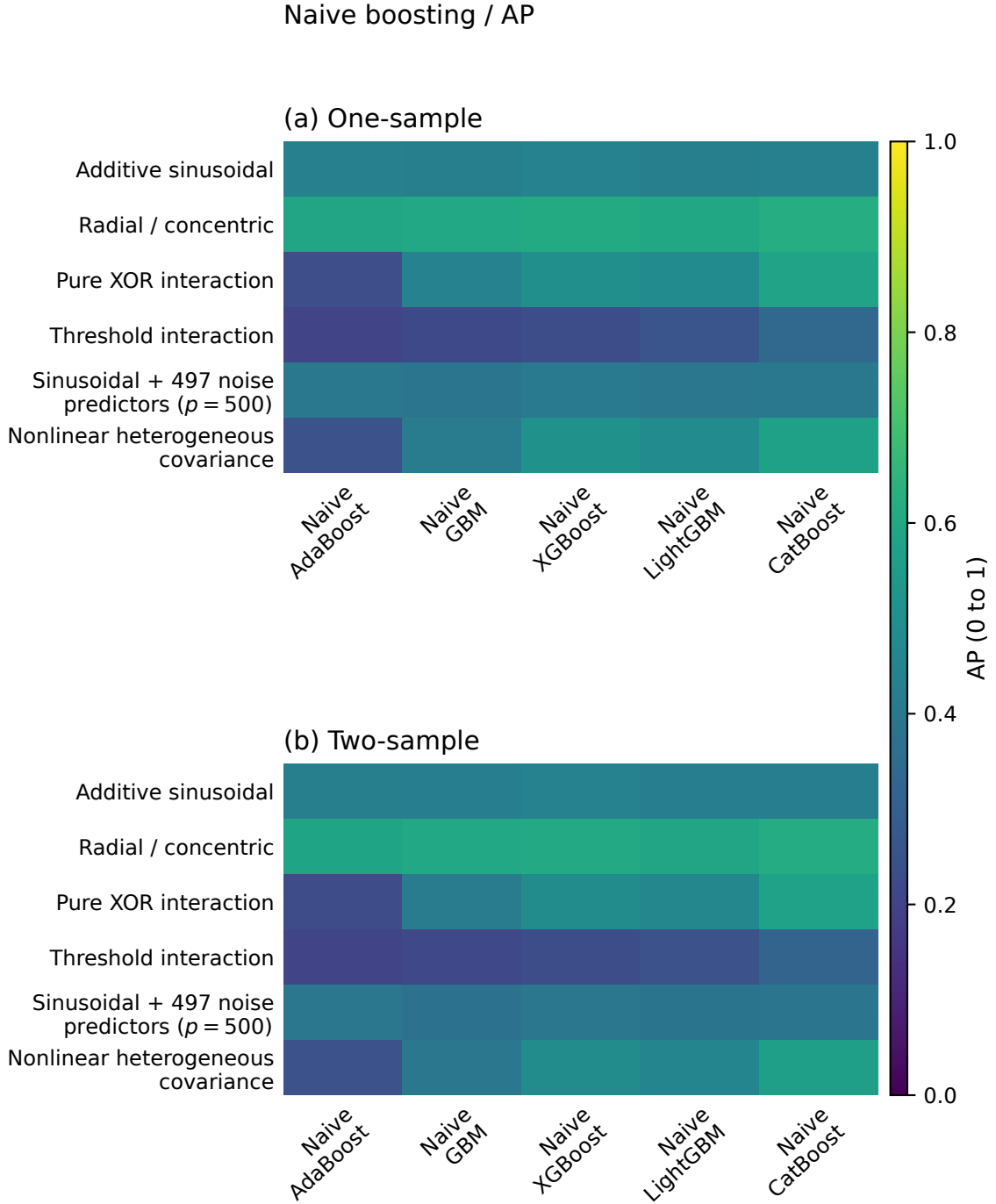

Naive boosting / AUROC
(a) One-sample
Additive sinusoidal
Radial / concentric
Pure XOR interaction
Threshold interaction
Sinusoidal + 497 noise predictors ($p = 500$)
Nonlinear heterogeneous covariance
Naive AdaBoost
Naive GBM
Naive XGBoost
Naive LightGBM
Naive CatBoost
(b) Two-sample
Additive sinusoidal
Radial / concentric
Pure XOR interaction
Threshold interaction
Sinusoidal + 497 noise predictors ($p = 500$)
Nonlinear heterogeneous covariance
Naive AdaBoost
Naive GBM
Naive XGBoost
Naive LightGBM
Naive CatBoost
AUROC (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0

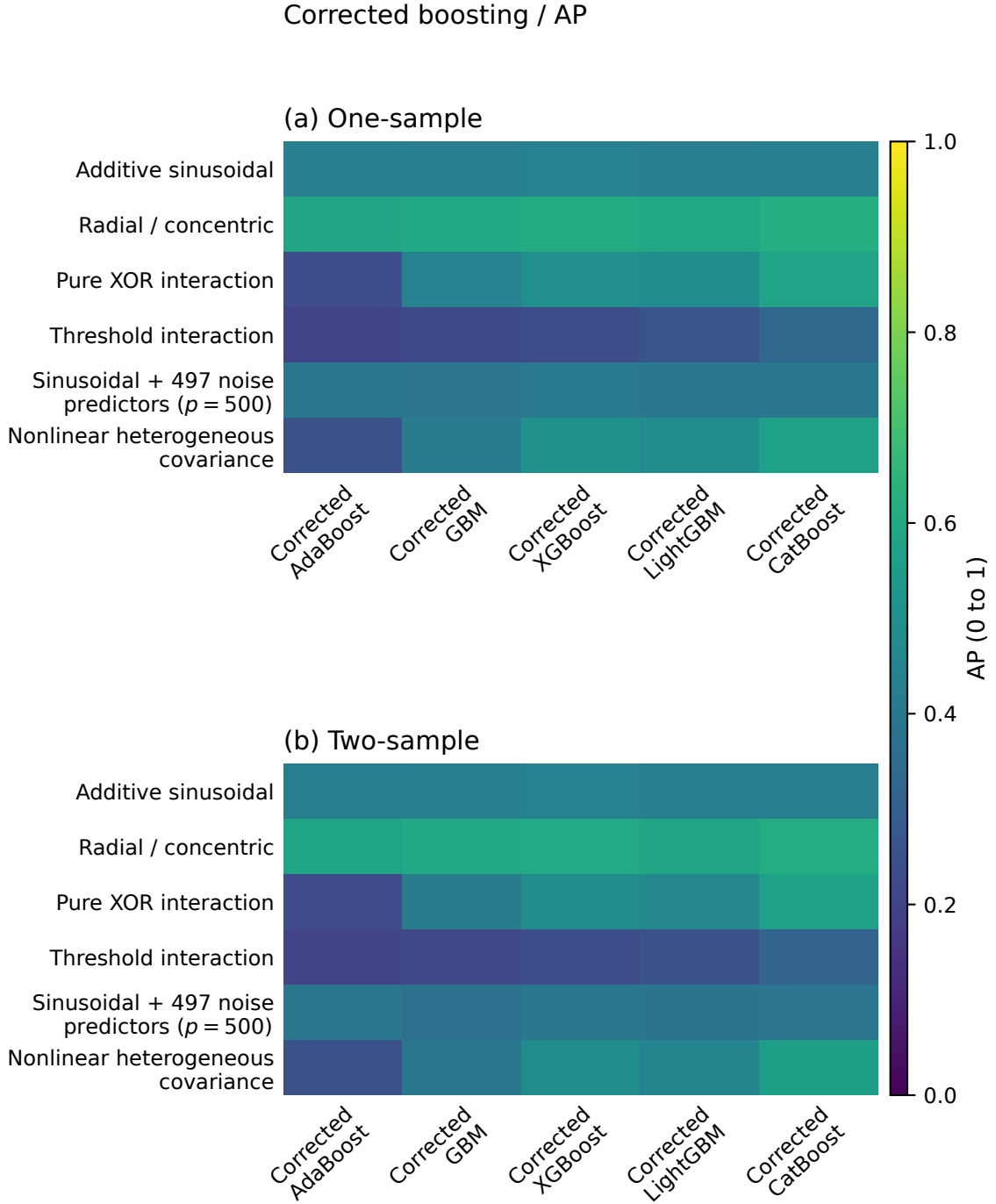

Corrected boosting / AUROC
(a) One-sample
Additive sinusoidal
Radial / concentric
Pure XOR interaction
Threshold interaction
Sinusoidal + 497 noise predictors ($p = 500$)
Nonlinear heterogeneous covariance
Corrected AdaBoost
Corrected GBM
Corrected XGBoost
Corrected LightGBM
Corrected CatBoost
(b) Two-sample
Additive sinusoidal
Radial / concentric
Pure XOR interaction
Threshold interaction
Sinusoidal + 497 noise predictors ($p = 500$)
Nonlinear heterogeneous covariance
Corrected AdaBoost
Corrected GBM
Corrected XGBoost
Corrected LightGBM
Corrected CatBoost
AUROC (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0

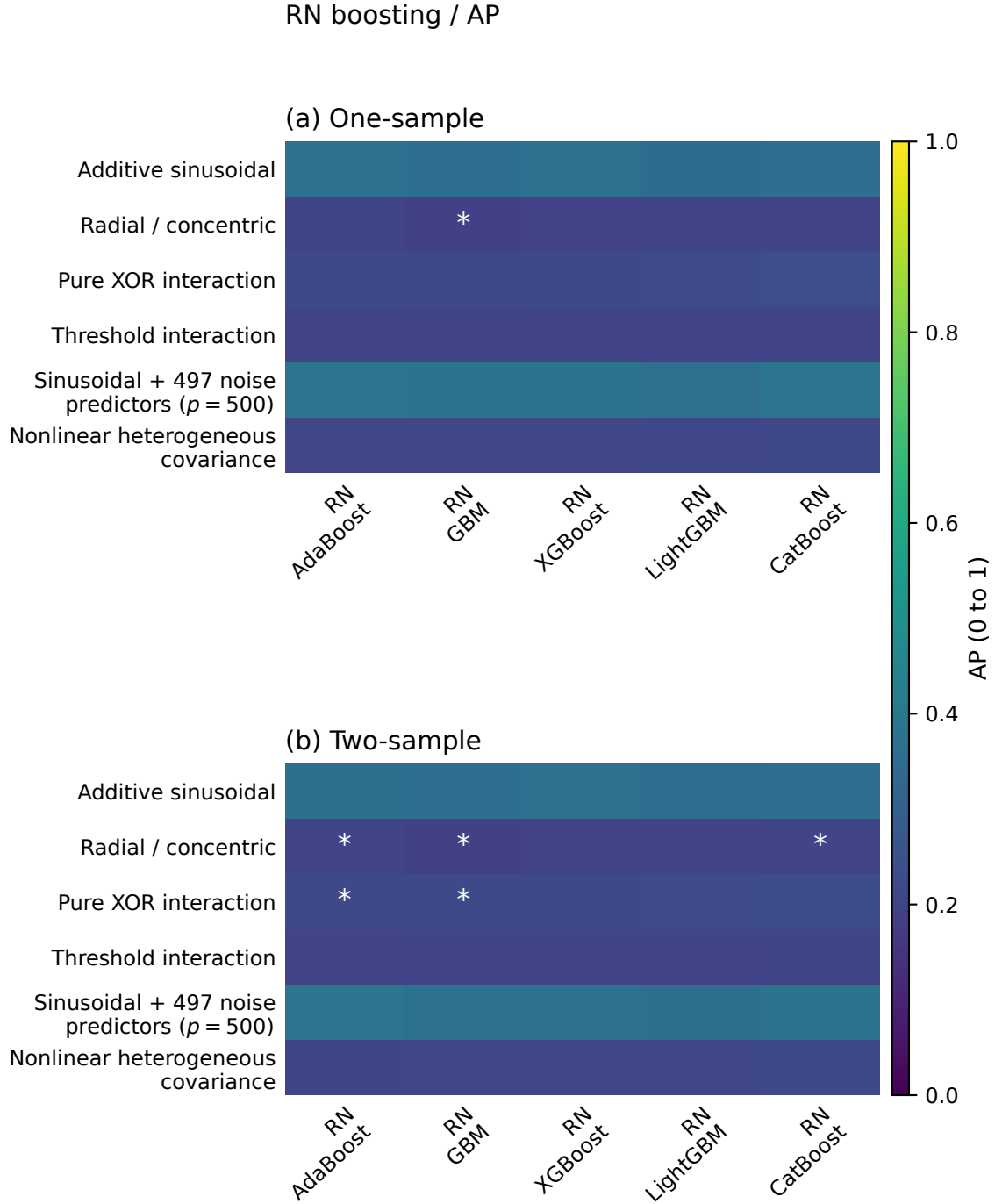

RN boosting / AUROC
(a) One-sample
Additive sinusoidal
Radial / concentric
Pure XOR interaction
Threshold interaction
Sinusoidal + 497 noise predictors ($p = 500$)
Nonlinear heterogeneous covariance
RN AdaBoost
RN GBM
RN XGBoost
RN LightGBM
RN CatBoost
(b) Two-sample
Additive sinusoidal
Radial / concentric
Pure XOR interaction
Threshold interaction
Sinusoidal + 497 noise predictors ($p = 500$)
Nonlinear heterogeneous covariance
RN AdaBoost
RN GBM
RN XGBoost
RN LightGBM
RN CatBoost
AUROC (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0

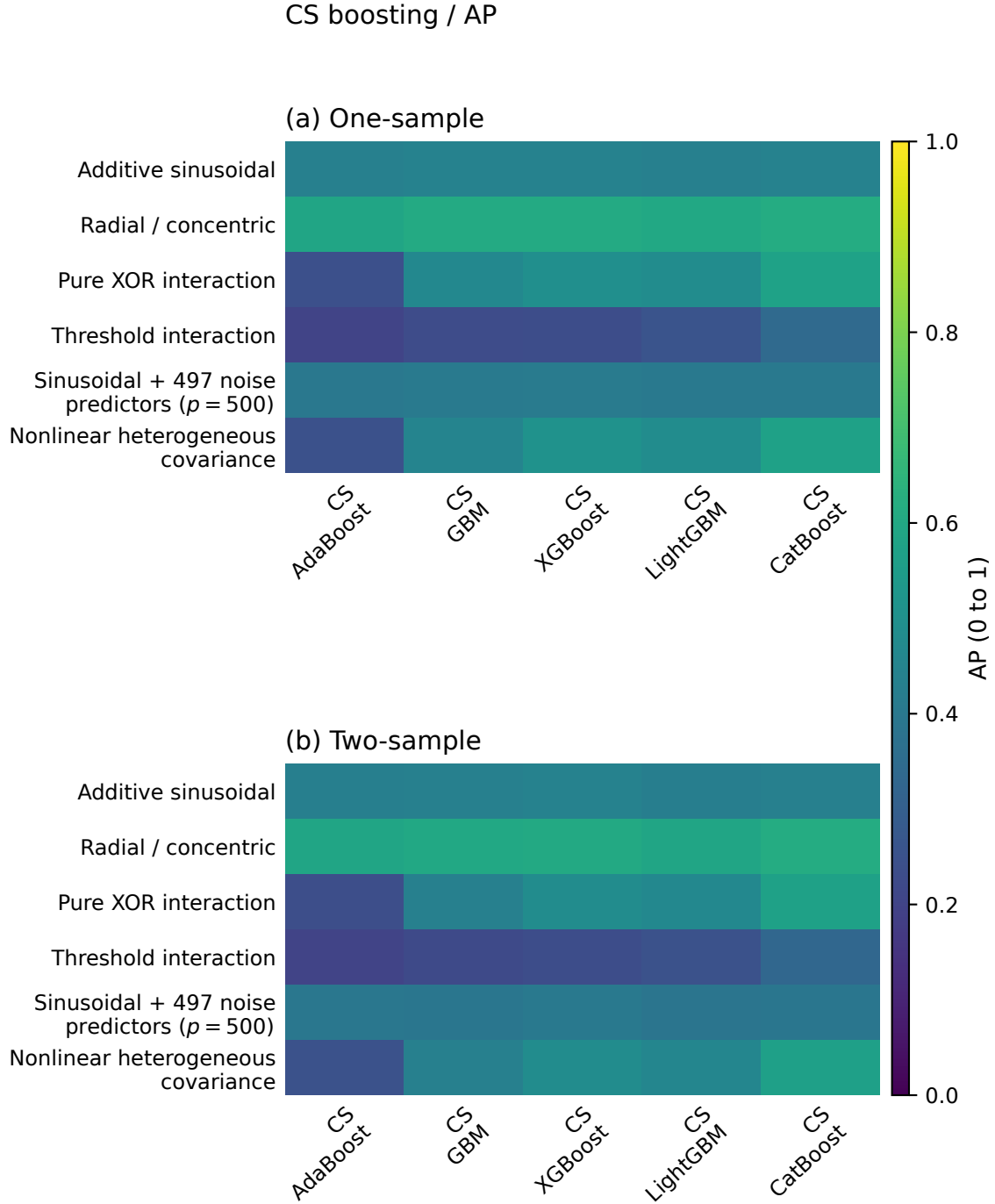

CS boosting / AUROC
(a) One-sample
Additive sinusoidal
Radial / concentric
Pure XOR interaction
Threshold interaction
Sinusoidal + 497 noise predictors ($p = 500$)
Nonlinear heterogeneous covariance
CS AdaBoost
CS GBM
CS XGBoost
CS LightGBM
CS CatBoost
(b) Two-sample
Additive sinusoidal
Radial / concentric
Pure XOR interaction
Threshold interaction
Sinusoidal + 497 noise predictors ($p = 500$)
Nonlinear heterogeneous covariance
CS AdaBoost
CS GBM
CS XGBoost
CS LightGBM
CS CatBoost
AUROC (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0


Nonboosting and native PU / AP
(a) One-sample
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uPU logistic
nnPU logistic
AdaPU
PU gradient boosting
(b) Two-sample
Additive sinusoidal
Radial / concentric
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Sinusoidal + 497 noise predictors ($p = 500$)
Nonlinear heterogeneous covariance
Logistic correction
uPU logistic
nnPU logistic
AdaPU
PU gradient boosting
AP (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0


Naive boosting / AP
(a) One-sample
Additive sinusoidal
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Pure XOR interaction
Threshold interaction
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Nonlinear heterogeneous covariance
Naive AdaBoost
Naive GBM
Naive XGBoost
Naive LightGBM
Naive CatBoost
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Naive GBM
Naive XGBoost
Naive LightGBM
Naive CatBoost
AP (0 to 1)
1.0
0.8
0.6
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0.2
0.0


Corrected boosting / AP
(a) One-sample
Additive sinusoidal
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